Chapter 17

Colour and Shade Matching in Restorative Dentistry

"The best and most beautiful things in the world cannot be seen or even touched – they must be felt with the heart".

—Helen Keller

Colour plays a critical role in the success or failure of esthetic dental restorations. Shade matching is as much an art as it is a science, and requires knowledge of colour science principles and the implementation of adequate shade-matching techniques. This chapter provides a working knowledge of the principles of colour and perception relative to understanding the complex nature of tooth colour and appearance. Shade-matching methods and tools for achieving predictable esthetic outcomes will be discussed along with resources available for continued improvement.

Colour and Perception

The Colour Triplet (Observer Situation)

The human perception of colour results from the interaction of three elements: *light source, object* and *observer* (Fig. 17.1). Because all three of these elements can be modified, any change in one element will affect the final perception of colour. The light source is a visible form of electromagnetic (EM) radiation that illuminates the object. When light strikes the object (tooth) a proportion of the energy is absorbed, transmitted or reflected. Colour perception is dependent upon the subjective ability of the human visual system to combine and interpret the physical interactions of light and object. The quantity of reflected light reaching the observer's eyes stimulates a subjective sensation in the brain that we experience as colour. In other words, the perception of colour ultimately resides in the brain and not merely in the property of the object. For this reason, colour can be defined as a psychophysical sensation provoked in the eye by the visible light and interpreted by the brain.

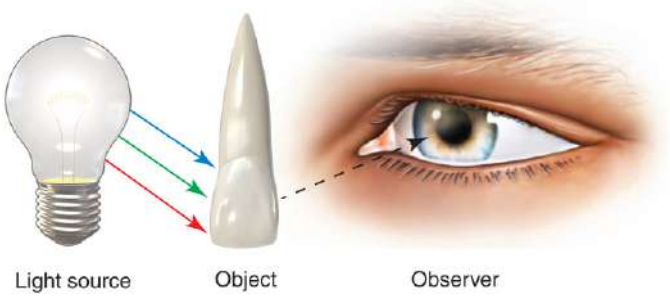


Figure 17.1 The colour triplet (observer situation).

Colour Vision

Rods and Cones

The human eye and brain, which enable colour vision, form an amazing and complex system. The visual system of a person with normal colour vision can identify millions of different colours.[1,2] At the innermost retinal layer of the eye are two types of specialized neurons that function as photoreceptors, called *rods* and *cones*.

I. Rods:
   i. The more numerous of the two photoreceptors are the rods, which are sensitive to low levels of light.
   ii. The rods are primarily responsible for our peripheral vision and are unable to detect colour.
   iii. In low levels of light, rods help us see objects in gray scale; as the light becomes brighter the rods become inactive.

II. Cones:
   i. The cones on the other hand operate in bright light and provide high-acuity colour vision. Both photoreceptors transform light into chemical energies that stimulate millions of nerve endings. The neural signals are transported by the optic nerve to the brain, where colour is interpreted.
   ii. There are three types of cones in the retina which are sensitive to different wavelengths of light: *blue, green* and *red*. The blue cones are most responsive to short wavelengths. The green and red cones are most responsive to medium and longer wavelengths, respectively, with some overlap.

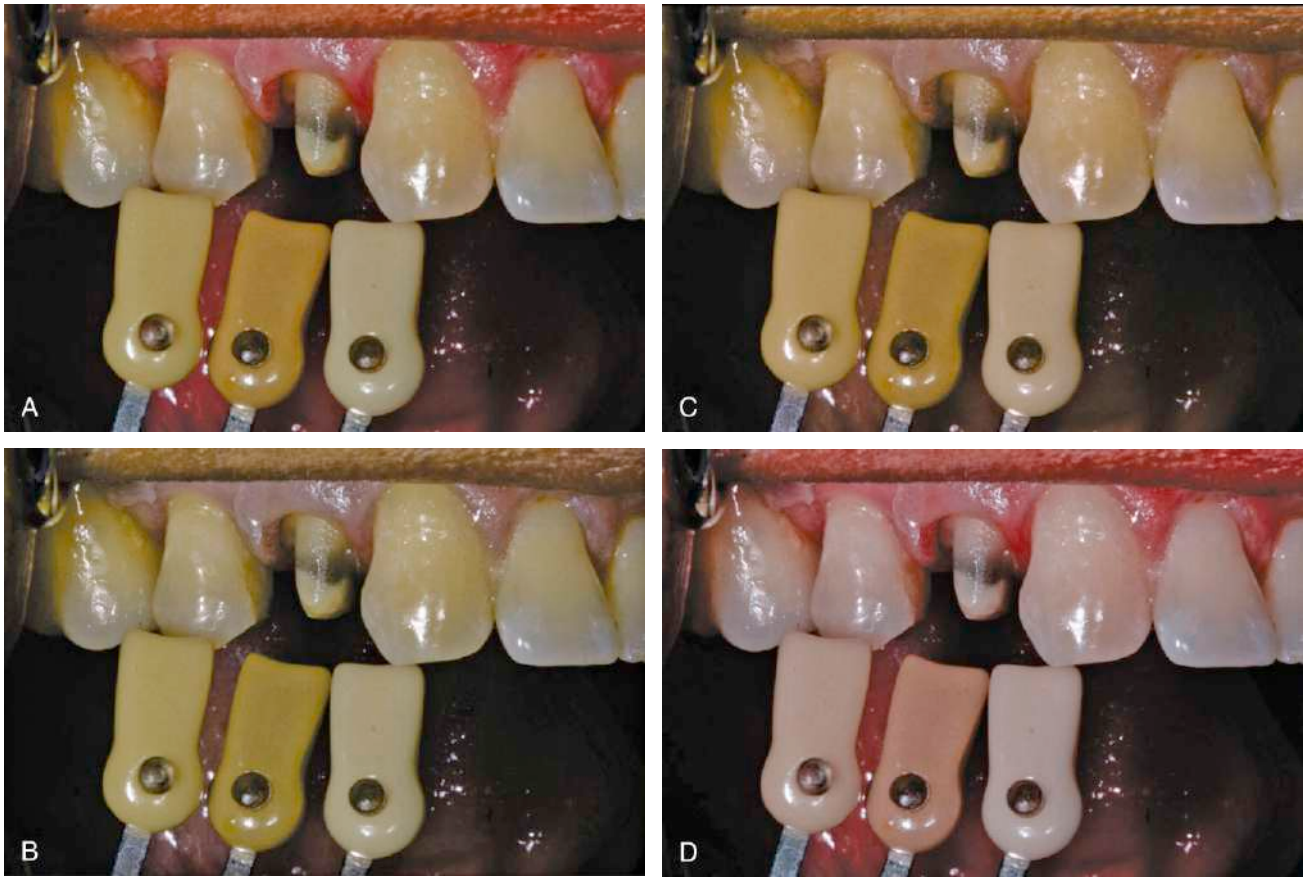


Figure 17.2 (A) Unaltered digital image, corresponding to normal colour vision (*upper left*), and after colour deficiency simulations. (B) Protanomaly (*upper right*). (C) Deuteranomaly (*lower left*). (D) Tritanomaly (*lower right*).
(The original image processed at http://www.color-blindness.com/coblis-color-blindness-simulator/)

Colour Deficiency

- Colour vision deficiency is a weakness or absence in one or more of the three types of cones. Individuals with these colour deficiencies still see colour; however, their colour vision is distorted (Fig. 17.2).
  - The most common colour deficiency in the population is an individual with a partial green defect known as *deuteranomaly*.
  - Other deficiencies are *protanomaly*, which is caused by a reduced sensitivity to red light.
  - *Tritanomaly*, which is someone lacking blue vision.
- Colour-blind individuals lack all three types of cones; this condition is called *monochromacy* or *achromatopsia*. Colour deficiency poses a challenge for the clinician when performing visual shade matching.
- Popular general tests to check colour vision are the *Ishihara test* (Fig. 17.3) and the *Farnsworth–Munsell test*.

Colour Dimensions

There are numerous systems and theories for arranging colour in some orderly fashion (i.e. colour spaces). The most popular system for visual colour matching in dentistry is based on the three-dimensional model devised by American artist Alfred H. Munsell in 1898.[2] Munsell's colour system forms the basis for the classification of coloured objects in the three dimensions: hue, value and chroma. It is important to grasp the concepts of hue, value and chroma to fully understand dental shade matching as described in this chapter.

Hue

Hue enables the distinction and differentiation among different colours. This colour dimension is the attribute by which an object is judged to appear red, orange, yellow, green, blue, purple etc. These are the *'pure'* colours found on a basic colour wheel or a simple box of crayons. These hues, which appear on the visual spectrum, are placed on a continuous, circular scale as seen in Fig. 17.4. Compared to a standard, the hue of an object would be communicated in terms such as redder, yellower, greener or bluer.

Value

The dimension of value refers to the lightness of a colour. It is the achromatic vertical scale from black to white representing all shades of gray. It is usually communicated in

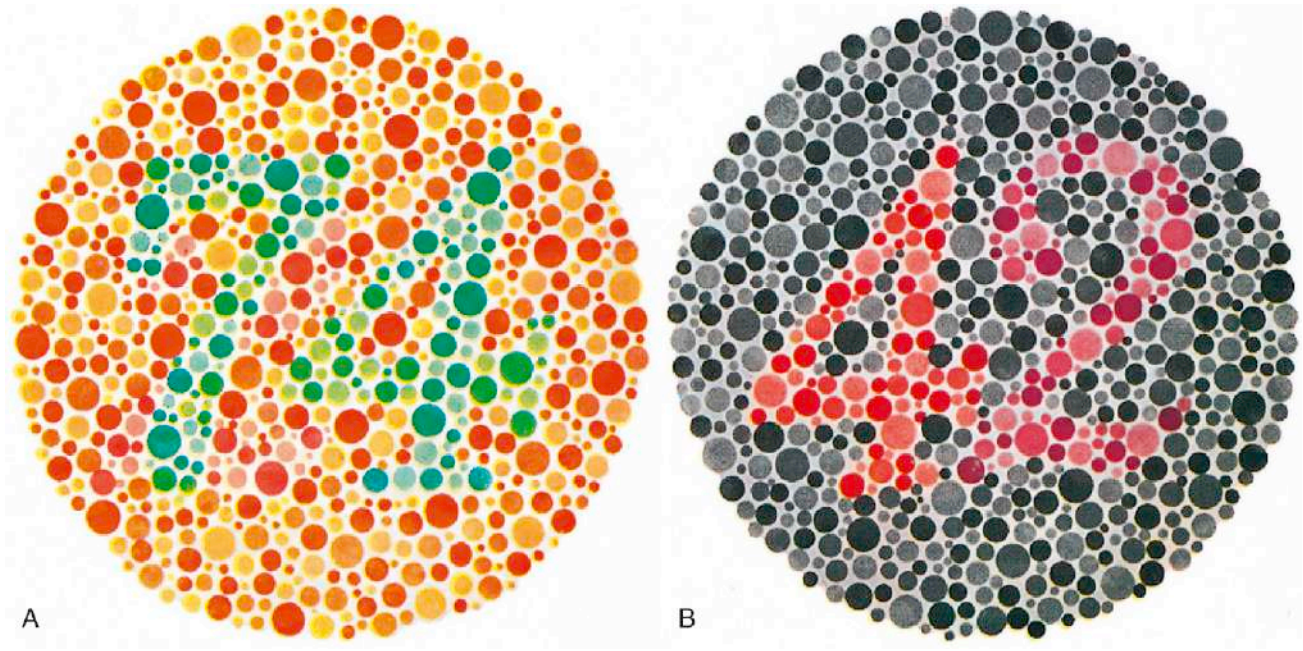


Figure 17.3 Ishihara Test. Observer should see number 74 (A) and number 42 (B).
(Reproduced from *Ishihara's Tests for Colour Blindness*. Tokyo, Kanehara & Co., but tests for color blindness cannot be conducted with this material. For accurate testing, the original plates should be used.)

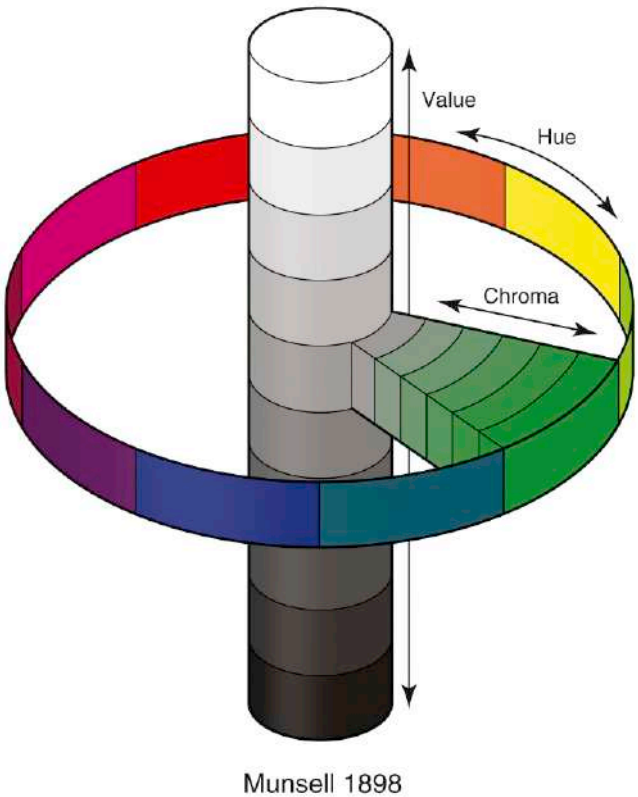


Figure 17.4 Munsell colour system with achromatic vertical value axis. Hues (*different* colours) are arranged in a 360-degree circle. Chroma is represented as the distance from the value axis, that is, the strength of the *same* colour increases as it departs from the value axis.

terms of lighter or darker. A tooth that appears lighter or 'brighter' as a result of bleaching would display an increase in value.

Chroma

Chroma is related to variation in strength of the same colour. The further away from the achromatic vertical axis, the higher the chroma (stronger, more intense). The closer the colour is to the achromatic (value) axis, the lower the chroma (paler, weaker). Chroma is often described as more chromatic or less chromatic. A tooth with a redder and/or yellower appearance at the cervical region, as commonly seen on a canine, can be described as more chromatic at that region. As the chroma increases, the hue becomes more specific.

Other Optical Properties

When light waves strike the surface of an object, the change in refractive index can cause waves to be reflected, absorbed or transmitted by the material. The combination of light speed and directional changes of the wave results in particular optical appearances of teeth as described later in the chapter.

Translucency

Translucency is the degree to which an object scatters light upon transmission, resulting in an appearance between complete opacity and complete transparency. Complete opacity will obscure the substrate beneath it by blocking the passage of light, while a completely transparent object will transmit

light without scattering and will clearly show the substrate beneath it.

Iridescence

Iridescence is a rainbow-like effect caused by the diffraction of light that changes according to the angle from which it is viewed or the angle of incidence of the light source. Iridescence occurs when light is diffracted from a thin layer that lies between two mediums of different refractive index (e.g. air and water), as in a soap bubble or a thin film of oil on water. Teeth do not display the property of iridescence, which is often confused with opalescence.

Opalescence

Opalescence is a milky iridescence that resembles the internal play of colours of an opal. In a natural tooth, opalescence is caused by light scattering between two phases of enamel that have different indexes of refraction. Short wavelengths of light are reflected displaying a blue hue, whereas longer light wavelengths, such as the orange and red, are transmitted through the tooth.

Gloss

Gloss is an attribute of visual appearance that originates from the geometrical distribution of light reflected by surfaces.[3] Particularly, gloss is a term used to describe the relative amount of mirror-like (specular) reflection from the surface of an object. Metals are usually distinguished by stronger specular reflection than that from other materials, and smooth surfaces will appear glossier than rough ones.

Fluorescence

Fluorescence is a form of luminescence, that is, a form of light emission by a substance as the result of some external stimuli. Following the excitation by light, usually ultraviolet (UV), a fluorescent substance will re-emit some of the absorbed energy in the form of longer wavelengths. When the luminescence continues after the source of excitation has been removed, the 'after-glow' is referred to as *phosphorescence*.

Surround Effects and Blending

Colour not only is defined by its colour dimensions and optical properties, but also depends on the surroundings in which the object appears, the adaptation of our eyes and our recent visual experience.[3,4]

Chromatic Induction

When two objects of the same colour are surrounded by different coloured backgrounds, an illusory sensation of colour can be created without direct stimulation of the corresponding cones. The two objects can be of the same colour when viewed in isolation, but when each is combined with different surrounds, the objects can have a perceived colour difference in relation to each other. This is what colour scientists call *chromatic induction.*

Contrast and Assimilation

Chromatic induction can generate either a contrast effect or an assimilation effect. When the object's colour shifts towards the complementary colour of its surroundings, this is known as *simultaneous colour contrast* (Fig. 17.5A). When the opposite occurs, that is, the perceived difference between the object's colour and its surrounding is reduced, this is known as *chromatic assimilation*.

Blending Effect

Clinically, the perceptual phenomenon of assimilation occurs when a restorative material (object) takes on the colour of the tooth (background/surround) and appears more similar combined than when viewed in isolation. In the dental literature, this visual blending of tooth and material is known as *blending effect* (BE).[5,6] The BE is commonly (and incorrectly) referred to as *'chameleon effect'.*

Complementary Afterimage

Similar to contrast and assimilation effects, our eyes can adapt to a recent visual experience and provoke illusionary perceptions of colour. When we stare at a solid colour for approximately 30 s or more, our photoreceptors become sensitized (retinal fatigue) and can create illusionary

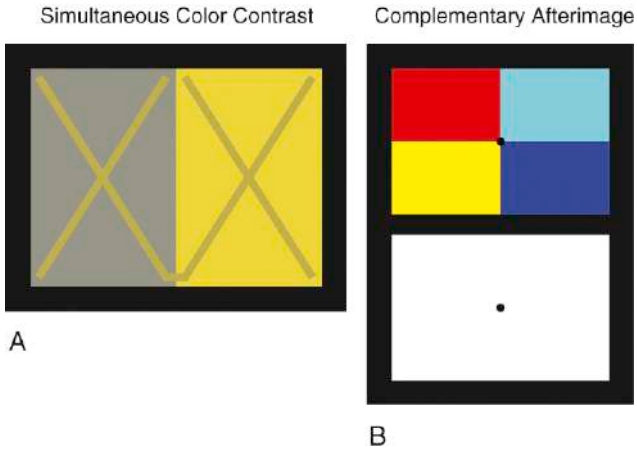


Figure 17.5 (A) Simultaneous colour contrast. The two Xs appear to be of different colours due to different backgrounds; however, when observed at their intersection at the middle base of the image it becomes obvious they are of the same colour. (B) Complementary afterimage. Stare for 30 s at the point where the four colours intersect; then switch gaze to the white target below the colour images. Retinal fatigue results in the illusionary appearance of the complementary colours.

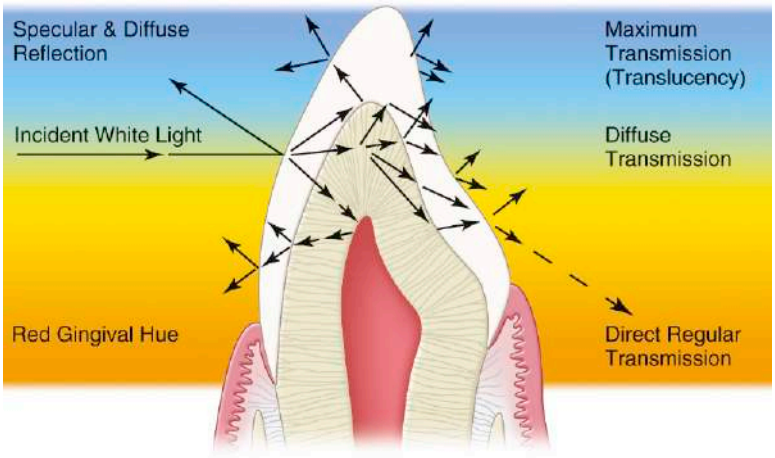


Figure 17.6 Complex interactions of light and tooth appearance.

images of the complementary colour; this is known as *complementary afterimage*. If one concentrates on a solid colour red target, for example, the red cones gradually respond less strongly to that reflected red signal. If one switches his or her gaze to a solid white target, now all colours are reflected to the retina and cones will send a strong green signal and a strong blue signal, but a weak red signal. One will see a cyan colour afterimage, cyan being the complementary colour of red (Fig. 17.5B).

Colour and Appearance of Teeth and Dental Materials

Tooth Colour and Appearance

How various reflections and transmissions of light integrate to generate perceived colours in human teeth is a complex process that is not entirely understood. The polychromatic appearance of the tooth originates from the relative interactions of light signals and perceptions (Fig. 17.6).

The quantification of tooth colour has been reported in the past by measuring the three regions of the tooth; *cervical*, *middle* and *incisal*.[7–9] Describing tooth regions can help understand how they are related to overall tooth colour and appearance. However, this approach is not necessarily sufficient to capture the polychromatic and complex tooth appearance (Fig. 17.7).

Dentin

In general, the colour of the tooth is not uniform. Dentin contributes significantly to tooth colour. This is particularly noticeable in the cervical region, where only a thin layer of enamel exists. This region is typically the most chromatic (Fig. 17.8), with the chroma progressively decreasing through the middle to the incisal third, displaying hues ranging from yellow to red.[7] Dentin is also the primary source of tooth fluorescence.[10]

Enamel

In a healthy unworn tooth, the incisal third is typically all enamel. If the enamel were isolated from the dentin, it would appear primarily achromatic like transparent or



Figure 17.7 Polychromatic appearance of natural teeth. Notice the progression of colour from cervical to incisal on teeth #8 and #9—from chromatic to translucent, changes in hue and incisal opalescence.



Figure 17.8 The exposed dentin showing accentuated chroma.



Figure 17.9 Translucency of isolated enamel.
(Courtesy: Adam J. Mieleszko, CDT.)

white frosted glass (Fig. 17.9). The translucency and value of the enamel can vary depending on many factors such as its thickness and age. Thick enamel generally appears higher in value relative to thin enamel. High-value white patterns, or white spots, may also demonstrate hypomineralized regions within the enamel. For anterior teeth, the enamel gets thinner towards the incisal and can appear gray to bluish against the dark background of the oral cavity. Depending on the transmission or reflection of light at the incisal edge, the incisal third may display an opalescent pattern with a distinct line of reflection described as the incisal halo (Fig. 17.10).

Colour-related Properties of Restorative Materials

Restorative materials should display colour and optical properties similar to those of dentin and enamel. In this section, we will primarily focus on composite resin and ceramic dental materials as they relate to colour compatibility, stability and interactions.



Figure 17.10 The incisal halo: distinct line of opalescent reflection at incisal edge.

Compatibility

When choosing dental materials such as composite resins or dental ceramics for restorative procedures, the shade selection of the material will be dependent on the brand or system in use. Most commonly a material is keyed to a commercially available shade guide. Shade guides will be further discussed under colour-matching tools. At other times, a restorative material may be labelled with a shade descriptor, such as 'universal dentin' or names like 'milky white' or 'pearl frost', without any reference to a shade guide. In either case, a material will be most compatible with a tooth when it has shades that mimic both dentin and enamel.

Stability

The colour stability of dental materials is a significant concern for colour and appearance in restorative dentistry. *When comparing composite resins and dental ceramics, resins are less colour stable after aging.*[11] Over time, resins are susceptible to extrinsic staining from dietary exposures and intrinsic degradation of the inherent chemical components.[12] Resins also can shift colour upon curing. Generally, microfilled composite resins become lighter and less translucent upon curing while microhybrid composite resins become darker and more translucent.[13] Ceramic materials, while more stable in service, can vary by batch and still undergo colour shifts upon firing and glazing.[14]

Interactions

An existing colour difference between the restorative material and the tooth can be lessened with favourable colour interaction properties, such as *layering* and *blending*.

- *Layering is the essence of tooth anatomy as layers of enamel and dentin of different thicknesses interact creating a polychromatic appearance*. Colour of both enamel and dentin can change over time due to dietary habits or aging, respectively. Given that the task is to mimic nature, layering is equally essential for creation of tooth coloured restorations.
- *A* BE *or colour shift of a dental material, such as composite resin or dental ceramic, towards the surrounding tooth colour is a desirable property*. This BE decreases the colour difference between the tooth-material interface giving the restoration a more lifelike and natural appearance. The BE is predominantly related to smaller restorations, surrounded by hard dental tissues, such as composite restorations. It can reduce suboptimal shade matches due to operational error or lack of satisfactory match in the shade guide or restorative material. The veneers designed with 'contact-effect' margins before and after cementation are another example of BE (Fig. 17.11).[5]



Figure 17.11 (A) Laminate veneers designed with 'contact lens-effect' margins. (B) The blending effect of indistinguishable translucent margins of laminate veneers.

Colour-matching Tools—Dental Shade Guides

Dental Shade Guides

The standard colour-matching tool used in dentistry for visual shade matching is the dental shade guide. Dental shade guides are tab-based tools fabricated from ceramic, resin or some other form of plastic or acrylic material. The shade tabs are typically arranged according to some dimension of colour, but because of the complex polychromatic nature of natural teeth, a given shade guide system will only serve as a guide and not as an exact colour matcher. While dental shade guides exist for oral soft tissues and facial skin, the focus of this section will be on guides designed for shade matching a tooth during the dental restorative procedure.

Commercial Shade Guides

Ceramic Based

For direct restorative procedures, there are many shortcomings associated with the use of a ceramic-based tooth shade guide for dental shade matching, but it is the most logical starting point for shade matching as most composite resins are keyed to a commercially available ceramic-based system.

Vita classical A1–D4 shade guide (VITA Zahnfabrik)

The most popular of the ceramic-based commercial guides is the Vita classical A1–D4 shade guide (VITA Zahnfabrik). This 16-tab shade guide can be arranged according to the hue order (A–D arrangement [Fig. 17.12A]) or according to an apparent light to dark arrangement (Value Scale [Fig. 17.12B]).

Each tab has a number and a letter. According to the manufacturer, the letters represent one of the following hue groups:

A = reddish-brown
B = reddish-yellow
C = grey
D = reddish-grey

The number next to the letter on the tab label represents the chroma and value within each of the A–D groups*: 1 = lowest chroma, lightest and 4 = highest chroma, darkest*. Within this system, shade B1 is the least chromatic and lightest of the reddish-yellow shades, whereas B4 is the most chromatic and darkest of the reddish-yellow shades.

- One way to use this shade guide is to observe the most chromatic portion of the patient's tooth, usually the cervical region of the canine and select the best hue group. Next, the best shade within that hue group should be selected based on the closest chroma number.
- A second way to use this shade guide is to arrange the shade tabs from light to dark according to the so-called 'Value Scale'. With the most contrasted end of the shade guide against the teeth, one would slide down or up the scale and stop on the best match. The 'Value Scale' is helpful when monitoring changes in colour after bleaching. However, the change from one tab to the next varies and with inconsistent value shift.

Vita 3D-Master shade guide *(VITA Zahnfabrik):*

I. VITA released a shade guide in 2007 specifically designed for bleach monitoring—*Vita Bleachedguide 3D-Master* (Fig. 17.13). It has many advantages over the 'Value Scale', from much wider range and more uniform colour differences between the adjacent tabs, to the inclusion of very light shades that enable monitoring of whitening efficacy of initially light teeth (e.g. shade B1 before bleaching).[15,16] This scale includes 29 shades (15 tabs with 14 interpolations) with colour evenly distributed between tabs.
II. *Vita Toothguide 3D-Master*.
III. *Vita Linearguide 3D-Master* (Fig. 17.14). This is a user friendly shade guide, simplifying the shade-matching method. Tabs are marked using a number–letter–number code that correspond to value–hue–chroma, respectively. There are five light gray holders, containing tabs from groups 0 to 5 (created based on decreasing lightness), with groups 0 and 1 on the same holder. Group selection is facilitated by having the middle tab of each group (M2) in a dark gray holder for initial shade matching. No more than 7 tabs, arranged in linear order, are viewed at once with the Linearguide. Furthermore, the 3D-Master system has the smallest



Figure 17.12 (A) Vita classical A1–D4 shade guide: A1–D4 arrangement. (B) 'Value Scale' arrangement of same shade guide.

'coverage error' (it covers the colour of natural teeth the best).[17] Consequently, it has been reported that this shade guide was preferred and outperformed others.[18]

A major concern with using the ceramic-based shade guides such as the Vita classical A1–D4 for selecting a composite resin shade keyed to this system is that the actual composite shade may or may not match the original classical tabs. It should be noted that large colour differences exist among the composite materials of the same shade designations.[19] If a restorative system is not supplied with a shade guide, then using a ceramic shade guide may be used as a starting reference.

Polymer Resin Based

Proprietary shade guides supplied with a restorative composite resin system are often fabricated using the same restorative material. In other words, they are resin-based tooth shade guides. Initially, this is a good option because they will have the same optical properties as the restorative

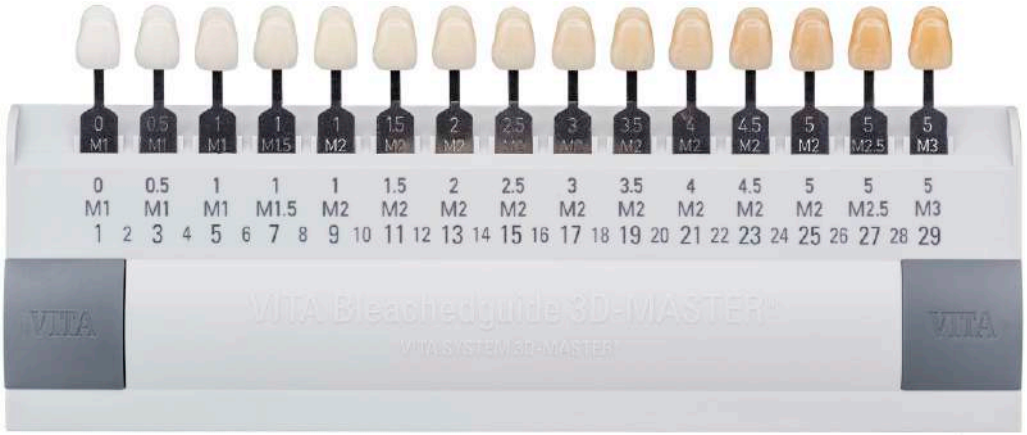


Figure 17.13 Vita Bleachedguide 3D-Master. Shade tabs are getting lighter and less chromatic (*right to left*), matching the changes that occur of natural teeth when exposed to whitening agents.



Figure 17.14 Vita Linearguide 3D-Master. Primary division into groups is made according to value (0–5); within the groups the chroma (1 = least and 3 = most) and hue (M = neutral middle hue; L = less red and R = more red) vary.

material. However, colour stability of the tabs may become problematic over time as the tabs are disinfected and get darker.

Other Materials (Plastics/Acrylics)

Shade guides are sometimes supplied with a restorative system fabricated from materials other than ceramics or resin-based materials, such as plastics or acrylics. These shade guides are generally inferior in comparison and are not good tools for shade-matching teeth.

Custom Shade Guides

Using the shade guide supplied with the restorative system in many cases would be advantageous over using a ceramic-based system. Yet, depending on the accuracy of the shade guide supplied with the system, it may be more accurate to use the actual composite cured on the tooth. It is important to fully cure the composite material when performing this evaluation as it has been shown that a shift in colour will occur. In general, restorative composites will shift upon curing to become less saturated. A more advanced alternative would be to periodically fabricate custom shade tabs from the actual restorative material using proprietary composite shade guides supplied by manufacturers (Fig. 17.15), moulds for fabrication of custom shade tabs (Fig. 17.16), a silicone index of a standard shade guide (Fig. 17.17) or cured pieces of composite placed onto respective areas of the tooth to be restored (Fig. 17.18).

Colour-matching Instruments

Colour differences can be difficult to verbally communicate between a tooth and a dental material. For example, what do we mean when we say a shade is too bright or too yellow? Digital instruments can be used to numerically quantify colour differences and to help arrive at an objective standard colour. Colour-matching instruments are designed to replicate how the human eye receives colour, yet without the subjectivity of the human visual system. Various devices have been used in dentistry, including digital cameras, scanners, spectrophotometers and colorimeters. Regardless of the device, the basis for all instrumental colour measurement is the hypothetical standard observer. Colorimeters, for example, receive

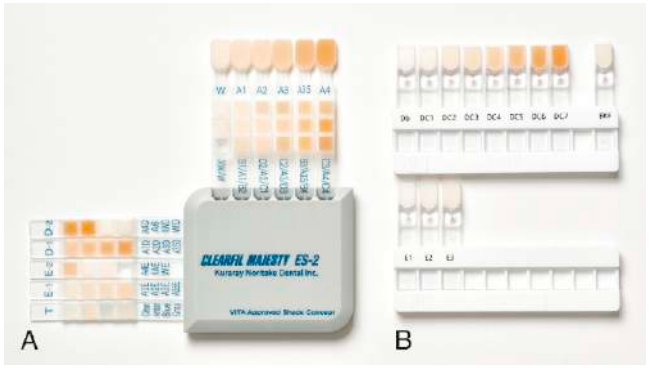


Figure 17.15 Proprietary composite shade guides supplied by manufacturers and made of monochromatic single-layered composites. (A) SDI shade guide. (B) Kuraray shade guide.







Figure 17.16 (A) Moulds for fabrication of custom composite resin shade tabs: (*1*) Tokuyama dentin and enamel moulds, tab holders and custom made tabs; (*2*) style Italiano mould for layered enamel and dentin, the enamel shell and a layered tab; (*3*) ultradent moulds for dentin, enamel or layered enamel–dentin, with respective tabs. (B) Dentin custom shade tab—clinical application. (C) Enamel custom shade tab—clinical application.





Figure 17.17 (A) Custom shade tabs from the actual restorative material made using a silicone index impression of a tab from a commercially available shade guide. (B) Clinical application: enamel shades UE1, UE2 and UE3 (*left to right*).



Figure 17.18 Cured pieces of composite placed on respective areas of the tooth to be restored can be helpful in shade matching.

and reduce light to its RGB (red-green-blue) components similar to that of the three colour receptors (cones) in the retina. The colorimeter expresses this information in terms of three values, converting the numeric information to a standard target colour. Similarly, spectrophotometers measure light reflected from the target relative to the total light. The amount of reflected light at each wavelength along the visible spectrum is calculated and translated into shade terminology that makes sense to the dental professional. Historically, dentistry has not been a large market for colour measuring devices. Many devices have come and gone without widespread adoption. The one instrument that has remained on the market over many generations is the Easyshade device manufactured by VITA. The latest generation of this spectrophotometer is the *Vita Easyshade V* (Fig. 17.19A). This instrument provides shade information correlated to the Vita



Figure 17.19 (A) Vita Easyshade V clinical spectrophotometer (*top*) and the instrument display (*bottom*). First, to measure colour of natural teeth, the tooth symbol (*the upper icon*) should be selected; then best matches in Vita 3D-Master (*left*) and Vita classical A1–D4 shade guide (*right*)*;* next, colour difference (ΔE*) and value, chroma and hue differences compared to selected shade and last, best matching tabs from Vita 3D-Master (*left*) and Vita classical A1–D4 (*right*) shade guides in cervical, middle and incisal third. (B) Vita Assist is a software for Easyshade V that enables recording, storage, editing and commenting of patient data and images. It facilitates communication on tooth shade and translucency between dentist and patient and between dentist and dental technician, including the communication via smartphone (Vita mobileAssist app).

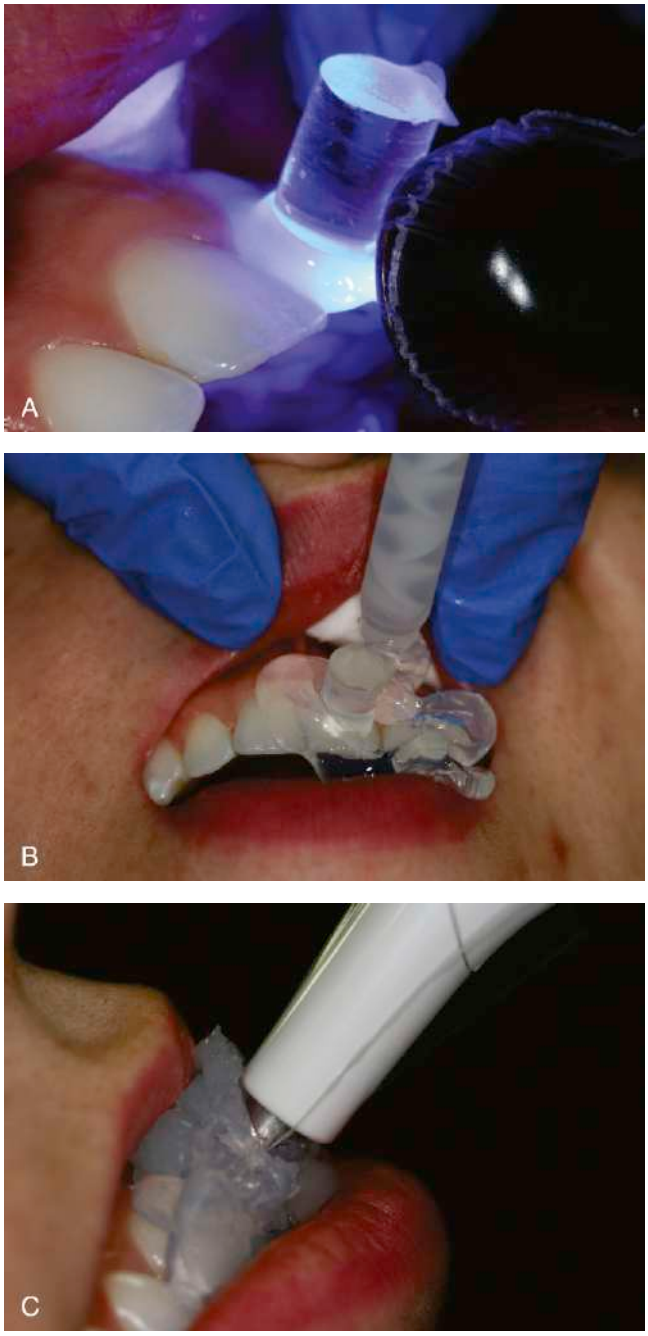


Figure 17.20 Fabrication of clear silicone individual positioning jig for repeated spectrophotometer measurements. (A) Acrylic rod cured to the tooth. (B) Injection of clear silicone material. (C) Measurement through the cylindrical tunnel created upon the removal of the acrylic rod.

classical A1–D4 and Vita System 3D-Master shade guides. Its proprietary software, Vita Assist, and smartphone application, Vita mobileAssist, allow Bluetooth transfer of the colour information and digital images to a computer for patient and laboratory communication (Fig. 17.19B). An advanced method used in research where repeatable measurements are necessary to monitor colour changes involves using a custom jig to avoid angulation of the probe and ensure precise measurements (Fig. 17.20).[20]

Table 17.1 Basic setting of DSLR camera for dental photography[19]

Manual mode	M (manual)
Focusing	Manual
Aperture	f22
Shutter speed	1.125 s
ISO	100
Image size	Maximum
Image format	JPG fine
White balance	Flash or 5300 K
Image optimization	Set to neutral
Flash output	One-half or one-quarter power (non-TTL flash)
Gridlines	On (for easier orientation)

While colour measuring instruments continue to improve, they still do not replace the operator. Instead, colour-matching instruments provide the dental professional with an objective tool to confirm a 'best match' among various shade guides. Ultimately, the best colour-matching tool would be the one whose results correspond to a clinician's normal colour vision.

Dental Photography

The use of photography is integral to a complete analysis of tooth colour and appearance. Digital single-lens reflex (DSLR, recommended) cameras, portable cameras and cell phones are used for digital photography. A DSLR camera can be used with a general-purpose lens or dedicated macro lens, and a variety of flashes: *single-point, ring, dual-point* and *twin tube flash*. Digital images can be captured and saved, probably most frequently in JPG format (with an adjustable degree of compression and consequently image quality), while advanced users typically prefer RAW format (with minimally processed data). Recommendations for basic settings of DSLR cameras for dental photography are provided in Table 17.1.[19] Mirrors and retractors are important accessories for digital photography.

Photography enables the use of a colour monitor to magnify a tooth image and discriminate subtle transitions of colour, use black-and-white contrast adjustments to study translucency patterns, as well as an essential tool for communicating colour to the dental laboratory technician for indirect restorations. Today, a photo can be taken of a tooth with a shade tab held on the same plane, along with a standard black/white/gray standard card, and software can provide a 'Colour Map' of the target tooth. One example of this type of dental shade-matching software is *Shade Wave* (Issaquah, WA). An attachment arm (Shade Arm) is available for fixing the distance, angle and position of the colour reference in the photo. The software will mathematically normalize, or colour correct, the image to

compensate for colour imbalances that occur when the image is taken and will cross-reference the standard card to generate a 'Shade Map', 'Value Map' and 'Translucency Map'. The value of this type of system is that it does not rely on any specific camera or camera settings, nor is it influenced by surrounding colours or light. Digital photography is frequently used in conjunction with general software programs (such as Adobe Photoshop and Corel Photo-Paint). Mobile apps are becoming increasingly popular for colour management.

Visual Shade-matching Method

Colour matching is dependent on controlling subjective variables such as colour vision deficiencies and eye fatigue, while at the same time controlling the proper viewing conditions and selecting the correct light sources. The following order of three preshade-matching steps and five shade-matching steps will provide a predictable method for visual shade matching.

Three Preshade-matching Steps

Step 1: Check Colour Vision

Prior to performing a shade-matching trial, it is imperative to screen the vision of the clinician for any colour deficiencies. A common myth about shade matching is that females are better colour matchers than males. While statistics show that up to 8% of males and 0.5% of women have a colour deficiency,[3] this does not equate to a gender superiority if both are trichromats, that is, have normal colour vision.[21]

Test for colour discrimination competency in dentistry consists of creating pairs of shade tabs from two identical shade guides (at least one set should not have original markings, such as A1–D4 of classical, visible).[22] Under controlled conditions, this test is an accepted standard for dental colour research but also could be implemented for everyday practice.

Step 2: Use Colour Corrected Lighting

The way we perceive colour is greatly influenced by the light source used to illuminate the object. In shade matching, we want to use an illuminant which best matches the white light of natural daylight. White light is a creation of our minds as a result of interpreting the spectral colours present in a particular illuminate; there is no 'colour white' on the visible light spectrum. White light is actually a mixture of all the colours of light. To understand correct lighting for shade matching we first should consider the issue of *correlated colour temperature* (*CCT*). It is based on early 20th-century German Nobel Laureate Max Planck's experiments in quantum physics. He developed a mathematical standard which correlates to a theoretical 'black body' whose spectral colour changes with temperature. The perfect black body emits the exact amount of electromagnetic radiation (light) as is absorbed. As the temperature increases, the object begins to glow a particular colour, turning 'red hot' near 1500 K. As the object gets hotter, it changes to a reddish-yellow around 3000 K (household incandescent lamp) and becomes 'white hot' near 5000 K as it moves throughout the visible spectrum, eventually turning blue-white. *For shade matching, light sources with a CCT of 5500 K and 6500 K are recommended*; the colours are correlated to standard phases of natural daylight.

The CCT of a light source alone is not the only specification important for correct lighting for shade matching. The *colour rendering index* (*CRI*) and intensity of the light source should be considered. The CRI is an average performance rating score for a light source based on comparisons to reference colours. The maximum score is 100 representing a full-spectrum light source (like the sun) that affects our colour judgment as natural daylight. For best colour rendering results, *light sources with CRI of 90 or higher are recommended*. The intensity of the light source, or *illuminance*, is measured in lux (lx) and should be between 1000 and 2000 lx. If the illuminance is too low, colours cannot be discerned; if the illuminance is too high, colours will be washed out. Despite using quality light sources with the correct CRI, CCT and intensity, some dental materials may have spectral properties that appear to match under a given set of viewing conditions and not match under another. This phenomenon known as *illuminant metamerism* at times is unavoidable due to the inherent properties of the dental material relative to the complex nature of natural teeth. It is advisable to use lights with different CCT to verify the existence of metamerism (Fig. 17.21).

There are a variety of lights that exhibit the recommended colour characteristics either as ceiling light, floor and table lamps or handheld lights. Handheld lights, such as *Rite-Lite 2 HI CRI* and *Smile Lite*, are becoming increasingly popular for visual shade matching. They can be used with

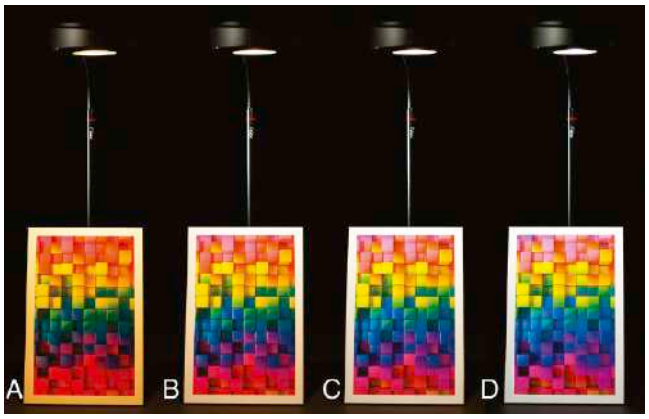


Figure 17.21 Influence of lights of different colour temperature on colour perception of the same object. (A) 3000 K. (B) 4000 K. (C) 5000 K. (D) 6500 K.

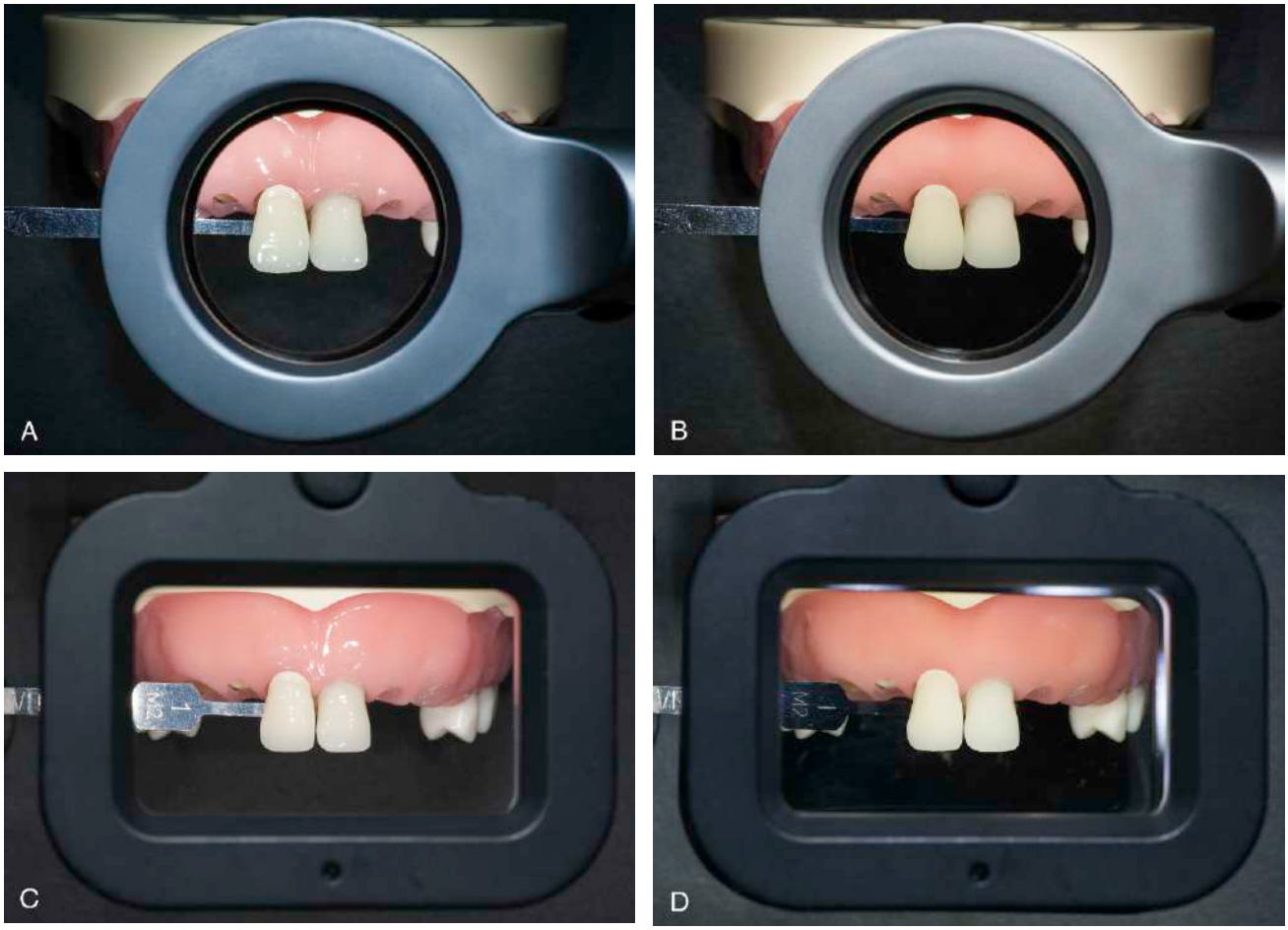


Figure 17.22 Handheld shade-matching lights. (A) Rite-Lite 2 HI CRI without polarizing filter. (B) Rite-Lite 2 HI CRI with polarizing filter. (C) Smile Lite without polarizing filter. (D) Smile Lite with polarizing filter.
(Photos from Clary JA, Ontiveros JC, Cron SG et al.: Influence of light source, polarization, education, and training on shade matching quality. *J Prosthet Dent* 116:91–97, 2016.)

or without (standard) polarizing filters (Fig. 17.22). Polarizing filters are intended to reduce glare, thus enabling better comparison of 'pure colour' and better visualization of colour transitions. However, no one normally observes a restoration and/or teeth through a polarizing filter, and the filters were not found advantageous in terms of improving shade-matching quality.[23] In addition, the Rite-Lite 2 HI CRI has three light options for shade matching: with CCT of 5500 K, 3200 K and their combination, which enables better visualization of metamerism.

Step 3: Control Surround/Viewing Conditions

As discussed earlier in this chapter (see Colour and Perception) the surrounding conditions can change the overall colour perception of an object. Any surround conditions that may influence colour perception should be addressed prior to shade matching. Distracting colour should be eliminated from the trial room with walls preferably painted a neutral gray colour. Any bold colours on the patient should be eliminated. Start by placing a neutral colour patient napkin over sparkling jewellery or bright clothes and having the patient remove reflective glasses and coloured lipstick (Fig. 17.23).



Figure 17.23 Patient is asked to remove intense lipstick prior to shade matching (note the lipstick on canine and premolar).

Five Shade-matching Steps

Once the preshade-matching steps have been addressed, the following shade-matching steps can be implemented to further improve visual shade-matching performance.

Step 1. Perform at the Beginning

The issue of timing is important to visual shade matching and should be performed at the beginning of the appointment. *Tooth dehydration* occurs during restorative field isolation and alters the tooth colour.[24] As teeth dehydrate, they will appear lighter, less chromatic and more opaque. Performing shade matching at the start of the appointment will also help prevent eye fatigue and strain on the visual system as time progresses.

Step 2. Set Light and Observer

This step involves using the *correct viewing distance* and *position* (Fig. 17.24). The correct viewing distance for shade matching will vary based on the clinician's visual acuity.

- Generally, the ideal distance should correspond to the best reading distance and *visual angle of subtense* (>2 degrees). For most, this distance will fall in the range of ***25–35 cm*** (10–14 in.).
- The angle of illumination and person performing shade matching relative to the shade tab is also an important factor, known as the *optical geometry*. The optical geometry of 45/0 degrees (light at 45 degrees, unidirectional, bidirectional or circumferential; observing at 0 degrees, perpendicular to tooth surface) or diffuse/0 degrees is typically used for clinical evaluation. The viewer's eyes should be on the same level as the shade tab.

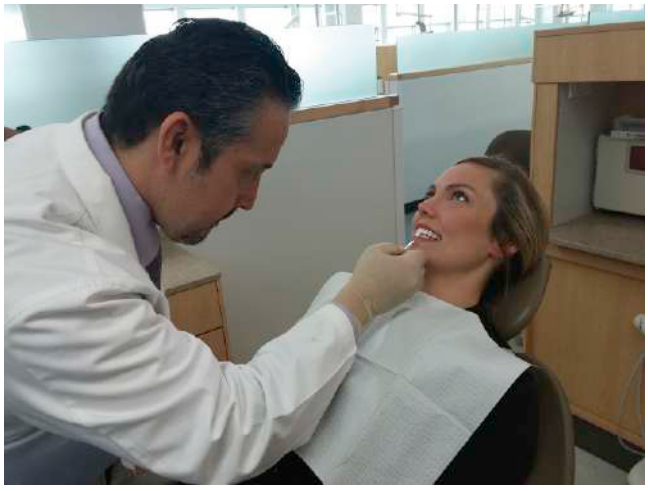


Figure 17.24 Shade matching should be performed at the distance of 25–35 cm (10–14 in.), with tooth being observed perpendicularly to its labial surface and with the 45-degree angle of illumination.

Step 3. Use Appropriate Technique

The shade-matching technique includes selection of the appropriate shade guide for the task (see *Colour-matching Tools—Dental Shade Guides* earlier in the chapter), appropriate shade tab positioning, technique and duration.

Shade Tab Positioning

After the position of the clinician is set, attention should be focused on the positioning of the shade tab. The tab should be placed in accordance with the clinical situation and the goal (overall tooth colour or colour in different regions).

The ideal position of the tab is on the same plane and the same vertical (cervical to incisal) orientation (Fig. 17.25). This is easy to achieve if the adjacent tooth is missing. However, this is most often not the case, so the shade tab can be placed in between upper and lower teeth, vertically or horizontally (Fig. 17.25) to the longitudinal axes of the natural teeth.

Shade tabs should not be positioned in front or behind the natural teeth (Fig. 17.26).The shade tab is sometimes placed in front of the adjacent tooth, but the shade tab will appear lighter solely for being physically closer to the eye of the clinician. One way to avoid this is to incline the shade tab to *approximately 120 degrees relative to the natural tooth* and observe both from a symmetrical angle (Fig. 7.25B). Attention should be given that the tab carrier is opposite the incisal edge as not to influence the evaluation of translucency.

As the middle portion of shade tabs, both cervical to incisal and mesial to distal, is the most accurate representation of the tab colour, tabs can be modified by reducing the cervical third or keeping only the middle third, as shown on Fig. 17.27. The tab ledges enable tab positioning at the same anterior–posterior plane with the tooth.

Macro-Mini-Micro Shade-matching Technique

Macro-phase

Shade matching typically begins with a quick glance and selection of potentially adequate tabs. The entire shade guide is used, positioned close to the tooth whose colour is being matched, and moved laterally to facilitate this selection (Fig. 17.28A).

Mini-phase

After this, the potentially adequate tabs are clustered together (in or outside the shade guide holder), and the best matches for incisal, middle and cervical third are identified (Fig. 17.28B).

Micro-phase

Once the best match or matches have been selected in the final micro-phase, one needs to determine and describe differences in hue, value and chroma between the regions of natural tooth and selected shades. Local colour characteristics such as white spots, amber stains, striation patterns and any other appearance attributes or details, including

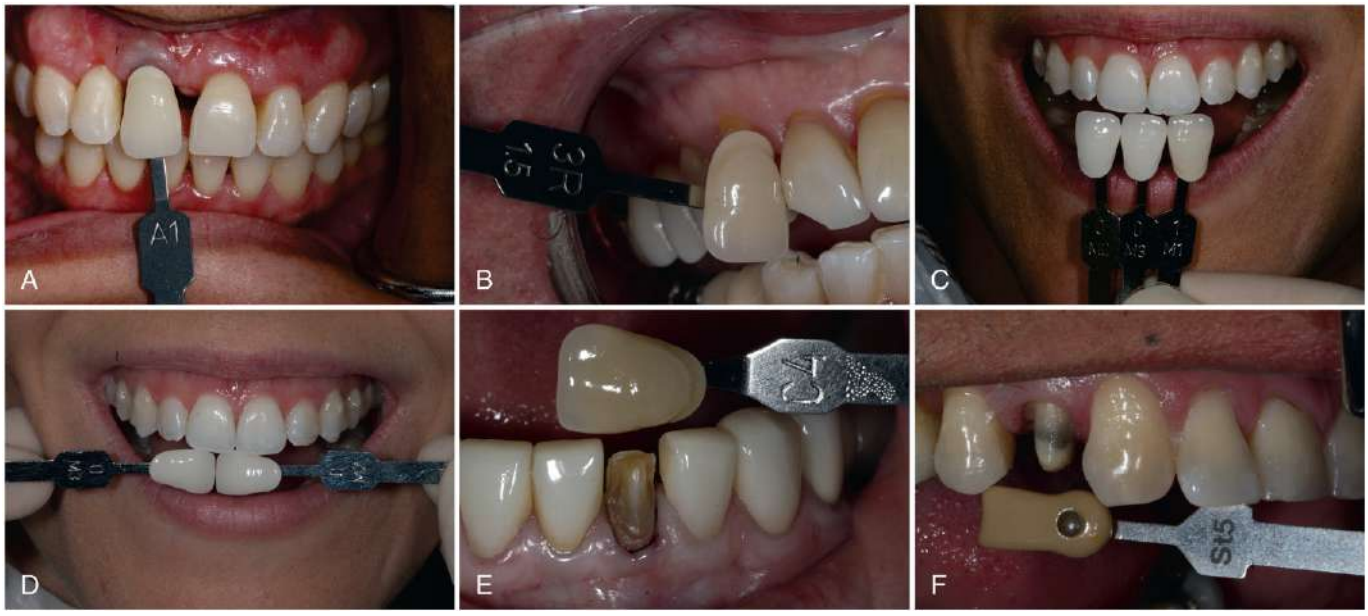


Figure 17.25 Suggested shade tab positions. Vertical positioning: (A) when the adjacent tooth is missing, the ideal position of the tab is on the same plane and the same vertical orientation (cervical to incisal); (B) when the adjacent tooth is present, the ideal position of the tab is inclined at approximately 120 degrees relative to the tooth, with both observed from the angle symmetrical and with the same vertical orientation (cervical to incisal); (C) on the same plane, incisal to incisal orientation. Horizontal positioning on the same plane, but at 90 degrees to tooth cervical to incisal orientation; (D) tooth shade matching; (E) stump shade matching with tooth shade tab and (F) stump shade matching with stump shade tab.

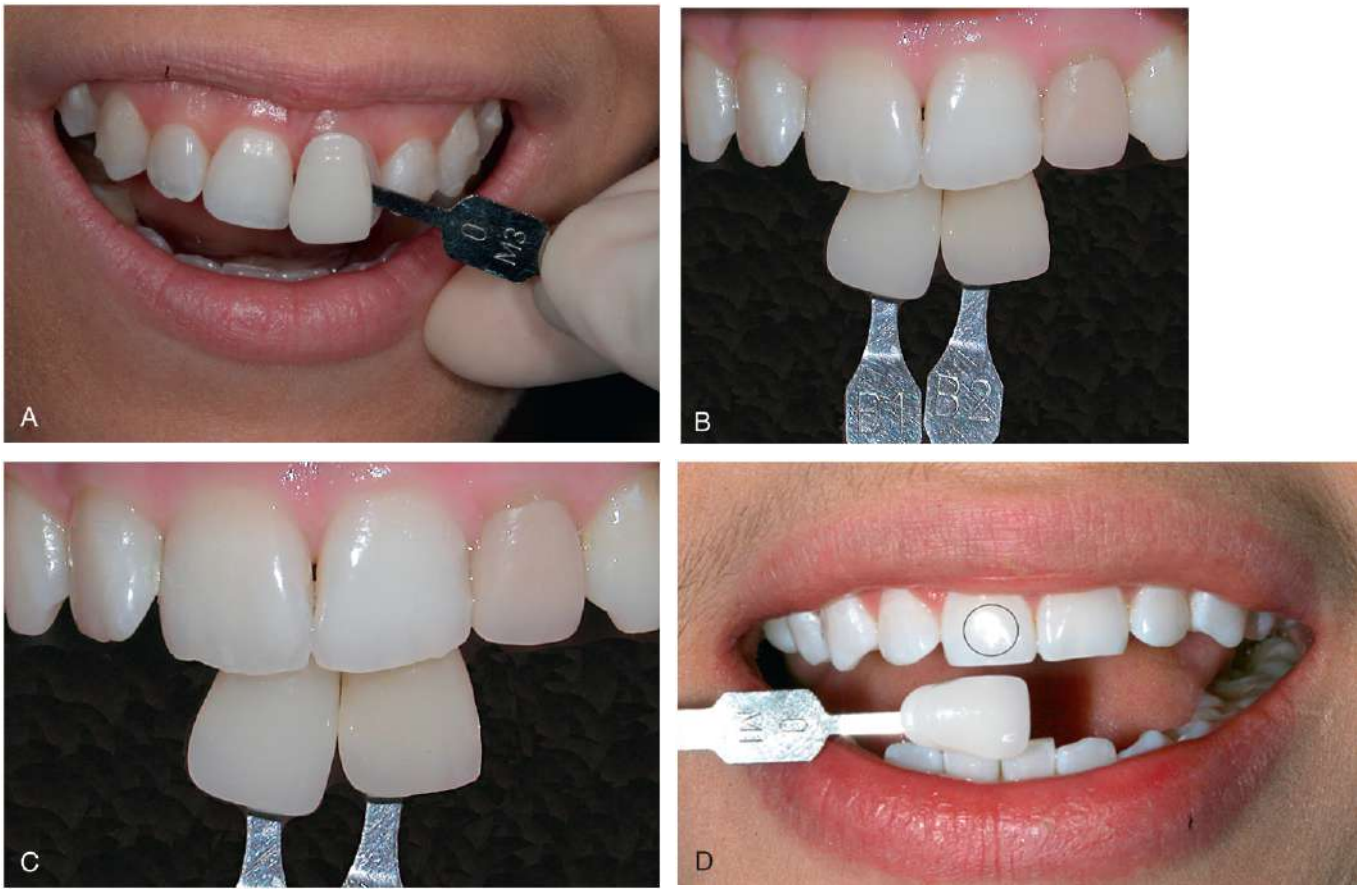


Figure 17.26 Suboptimal shade tab positioning and/or image capturing. (A) Tab in front of tooth appears lighter. (B) Tab behind tooth appears darker. (C) Shade designation not captured in image. (D) Light reflection of tab washes out shade.



Figure 17.27 As the middle portion of shade tabs, both cervical to incisal and mesial to distal, is the most accurate representation of the tab colour, tabs can be modified as follows: first, by reducing the cervical third: (A) shade tabs; (B) modified shade tab positioned on the same plane, cervical (*tab*) to incisal (*tooth*) and (C) modified shade tab positioned on the same plane, incisal (*tab*) to incisal (*tooth*). Second, by keeping only the middle third: (D) shade tabs; (E) tab on the same plane, underneath the natural tooth and (F) tab at 120 degrees relative to the natural tooth, observed/photographed from a symmetrical angle.

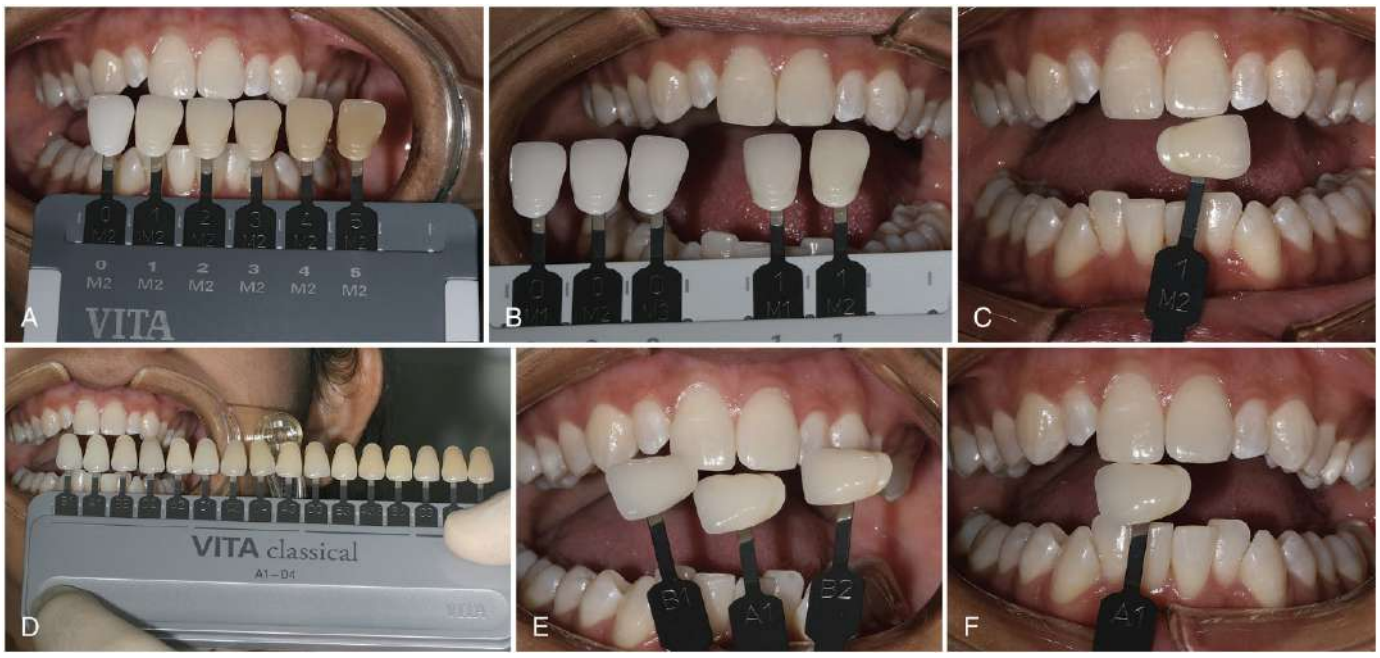


Figure 17.28 Macro-mini-micro shade-matching techniques. With Linearguide 3D-Master: (A) *macro-phase:* starts with the entire shade guide and a quick glance and selection of potentially adequate tabs (in this case, the group selection, 0–5); (B) *mini-phase*: the potentially adequate tabs are clustered together (in this case into groups) and the best matches for incisal, middle and cervical third are identified and (C) *micro-phase*: in the final phase, determine and describe differences in hue, value and chroma between the regions of natural tooth and selected shades. Local colour characteristics and other appearance attributes or details should be documented and described. The same also applies with Vita classical A1–D4, arranged according to 'Value Scale': (D) *macro-phase*: selection of potentially adequate tabs; (E) *mini-phase*: these tabs are clustered and the best matches by thirds are identified and (F) *micro-phase*: determining and describing differences in hue, value and chroma between the regions of natural tooth and selected tabs.

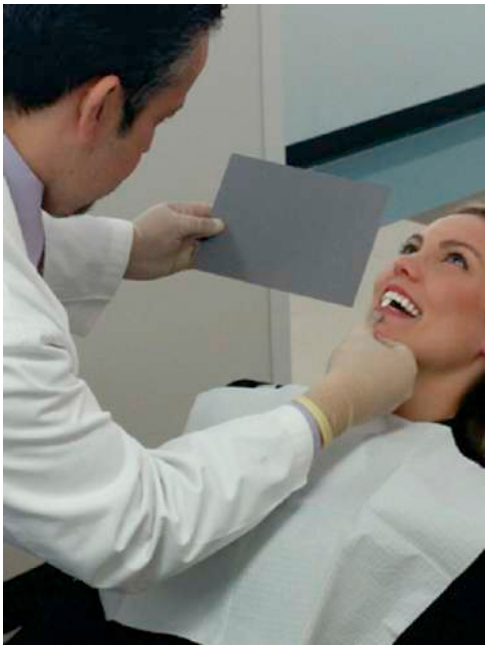


Figure 17.29 The clinician glances at a neutral gray card to rest his eyes in between shade-matching trials (lasting 5–7 s).

translucency, enamel cracks and craze lines, should be documented and described (Fig. 17.28C).

The same process is followed when using the Vita classical shade guide (Fig. 17.28D–F)

Shade Matching Duration

A key to discerning the difference in shade between two colours, especially if the colour difference is small, is to use very short glances for visual shade matching. This will help avoid illusory sensations of colour (chromatic induction and complementary afterimage) due to saturation of the rods and cones.

- *It is not advisable to stare at any colour during shade matching*. Staring at blue for example will create the illusion that the teeth are more yellow than they actually are in reality. It is preferable to see the true colour, not a complementary afterimage.
- *To avoid staring at the target, it is best to gaze at a neutral gray card in between shade-matching trials* (Fig. 17.29).
- Ideally the glances at the target shade should last only *between 5 and 7 s.*

Step 4. Communicate

The next step in the shade-matching process is to *capture, record* and *communicate* the selected tooth shades. The shades observed for direct restorations should be mapped out to assist in material selection during the restorative procedure. A simple sketch of the tooth illustrating the cervical, middle and incisal shades can help document the selected shades even through colour shifts after field isolation.[3] The aid of digital photography, software or data from a shade-matching instrument can nicely supplement visual shade matching in colour documentation and communication.

Step 5. Verify

The final step is to verify the colour of restoration (direct/indirect) visually, using several different lights, shade-matching distances, observation angles and/or observers. The same is true for verification using shade-matching instruments. The findings before the restoration is fabricated can supplement communication and case documentation. For direct composite resin restorations, the verification step can be made with a mock-up restoration or by simply applying (and light-curing) a small increment of the selected composite material to the area(s) of the tooth corresponding to the selected shade. The selection can then be fine-tuned accordingly. If this approach is used, the operator should consider variables such as viewing angle, proximity and others that may influence the results.

Improving Shade-matching Skills

Shade matching is a skill that can be learned and improved upon with implementation of sound colour principles. Like any skill worth mastering, it will require consistent and intentional practice of the methods covered in this chapter. The amount of time devoted to colour education and training in dental curriculums has been lacking in the past.[25] While up-to-date content is still lacking among educational institutions, significant advances have been made and greater resources are currently available for further study on the topic (Table 17.2).

Table 17.2 Available colour education and training programs

Name	Type	Publisher	Available at
CMC	Didactic and hands-on CE program	SCAD	www.scadent.org
DCM	Colour education and training software	SCAD	www.scadent.org
A Contemporary Guide to Colour and Shade Selection for Prosthodontics	Educational DVD	American College of Prosthodontists	http://dentistry.llu.edu/continuing-education/
TT	Colour training software	VITA Zahnfabrik	www.toothguide.com

CMC, Colour Matching Curriculum; *DCM*, Dental Colour Matcher; *SCAD*, Society for Colour and Appearance in Dentistry; *TT*, Toothguide Trainer.

The Colour Matching Curriculum (CMC) is the newest and most comprehensive colour education and training program for dental professionals and dental students.[26] This half-day continuing education program consists of didactic (colour concepts and resources, colour-matching conditions, methods, tools and communication) and hands-on (visual and instrumental colour matching) portions.

Summary

Understanding the science of colour is imperative to ensure accurate shade matching during esthetic dental procedures. Appreciation of the various optical properties of colour and using appropriate colour-matching guides would make this subjective art into an objective clinical reality.

Appendix

The Curious Case of a 'Bleaching Mishap'

A 22-year-old woman with 'Hollywood White' expectations presented with a chief complaint of 'two dark front crowns' (Fig. 17.30A). Her dental history reveals gingival



Figure 17.30 A patient with 'Hollywood White' expectations. The crowns matched natural dentition before at-home whitening, which appeared 'dark' upon whitening. (A) Preoperative smile. (B) Preoperative smile, retracted. (C) Shade tab 0M3 with black-and-white standardized card. (D) Shade tab 2M2 (stump shade) of prepared teeth. (E) Postoperative smile with all ceramic crowns on the upper right central incisor and upper left lateral incisor.

recession and overbleaching of her natural teeth with at-home whitening products resulting in a mismatch of all-ceramic crowns on the upper right central incisor and upper left lateral incisor (Fig. 17.30B). The patient declined treatment for gingival recession while accepting treatment to replace mismatched crowns. A ceramic-based shade tab (Vita Linearguide, 0M3) is used with a black-and-white standard card held on the same plane as the teeth to be restored. In conjunction with digital photography and the standard reference card, digital shade mapping with computer software can be performed for effective laboratory communication (Fig. 17.30C). The mismatched crowns were removed to reveal the natural high chromaticity of the dentinal preparations. The colour of the preparations, 'stump shade', is photographed for further laboratory communication with a reference shade tab (Vita Linearguide 3D-Master, 2M2) held in a horizontal position with the most chromatic portion of the tab (cervical) in contact with the tooth reference (Fig. 17.30D). All-ceramic crowns were delivered with improved colour matching to bleached natural teeth (Fig. 17.30E).

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